

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Matlab 2016a, Matlab 2017a, Matlab 2017b, ScanImage 2017, Fiji (https://fiji.sc/), FicTrac (http://rjdmoore.net/fictrac/)
Data analysis	All analyses of calcium imaging and electrophysiology data were performed using custom code written in Matlab 2017b (electrophysiology) and Matlab 2016b (calcium imaging) (see Methods for full description of analysis, code is available at https://github.com/wilson-lab/FisherLuDAlessandroWilson_AnalysisCode). Confocal images were analyzed using Fiji (ImageJ) and cell body counting was performed using the Fiji Cell Counter plugin (Schindelin, J. et al. Fiji: an open-source platform for biological-image analysis. Nat Methods 9, 676-682, (2012).)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	For genetic perturbation experiments (Fig. 3c), the number of experiments performed was determined by first collecting a pilot data set (n=4 for each of the 3 genotypes using the R20A02-Gal4 driver line). Based on the initial effect size, power analysis was used to determine the number of experiments needed to test the hypothesis that visually-evoked inhibition was smaller in the experimental genotype. For all other experiments, sample sizes were chosen based on standard sample sizes in the field.
Data exclusions	Figs. 1 & 2: Epochs were included if the cell was healthy; specifically, this meant that the epoch-averaged voltage was below -33 mV and within 15 mV of the voltage observed at the start of the first epoch of the experiment, and also if the spike amplitude was >50% of the amplitude observed in the first epoch. Closed-loop epochs were included if the fly visited all heading directions during that epoch. Cells were included if ≥ 2 open-loop epochs met these criteria; in Fig. 2 we also required that ≥ 2 closed-loop epochs met these criteria. Fig. 3: Cells were included if ≥ 2 open-loop epochs met our cell health criteria. A single recording from the UAS/+ control genotype was excluded because the biocytin fill showed that it was not an E-PG neuron. Fig. 4: 5/24 flies were excluded due to either weak fluorescence or an unstable offset between the angle of the E-PG bump and the fly's heading angle at the end of the initial closed-loop 1-cue epoch. Fig. 5: Cells were included if the epoch-averaged voltage from all epochs of the experiment (pre-training, training, post-training) was <-33 mV, and if the fly visited all heading directions during the 2 epochs (8 min) of 1-cue closed-loop prior to training and during the final 2 epochs (8 min) of 2-cue closed-loop training. We required that the fly's mean yaw velocity was >20°/s during the final 2 epochs of the 2-cue closed-loop training; 10 cells were excluded due to this restriction. We also removed recordings where the visual and/or heading turning curves were almost flat during the pre-training period (max-min ≤ 2mV); 6 cells were removed due to this restriction. On occasion, during E-PG neuron electrophysiological recordings, we observed unexpected large inhibitory postsynaptic potentials with a stereotyped sharp onset, a large amplitude (>15mV), and a stereotyped time course. They were followed by a prolonged period of depolarization when the variance of the voltage trace was also diminished. These events interfered with visual and heading tuning measurements, and so for Figs. 1-3, any epoch where such an event occurred was excluded from the analysis. For Fig. 5, the event was clipped but the rest of the epoch was used; 5% of open loop epochs and 10% of closed-loop epochs were clipped in this manner.
Replication	For genetic perturbation experiments (Fig 3c) we reproduced the effect of R neuron silencing with two independent driver lines (R20A02 and R54E12). For all other experiments, results were replicated in different individuals within each data set.
Randomization	For genetic perturbation experiments (Fig 3c) flies were grouped based on genotype (neuronal activity manipulated by Kir2.1 expression vs control genotypes). For comparison between trained and control flies (Fig 5) the experimental protocol that would be performed (control vs training) was always decided on prior to starting the experiment. No other randomization was performed.
Blinding	The experimenter was not blind to genotype except when genetic perturbations were used: Figure 3c (Kir2.1 perturbation). For the Figure 3c data set collected for driver line R20A02-Gal4 the experimenter was blind to genotype after the pilot phase; because Fig. 3c pilot data were indistinguishable from subsequent data, all data were ultimately pooled, and overall the experimenter was blind to genotype in 67% of these recordings. For the data set obtained using the driver line R54E12-Gal4, the experimenter was not blind to genotype because the experimental genotype was obtained at a lower-than expected (sub-Mendelian) frequency, making it impractical to blind the experimenter.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems	Methods
n/a	Involvement in the study
☐ ☒ Antibodies	☒ ☐ ChIP-seq
☒ ☐ Eukaryotic cell lines	☒ ☐ Flow cytometry
☒ ☐ Palaeontology	☒ ☐ MRI-based neuroimaging
☐ ☒ Animals and other organisms	
☒ ☐ Human research participants	
☒ ☐ Clinical data	

Antibodies

Antibodies used	-rat anti-FLAG (Novus Biologicals), Cat#: NBP1-06712B, RRID: AB_10006034, clone#: L5, lot#: B-3 -rabbit anti-HA (Cell Signaling Technologies), Cat#: 3724 RRID: AB_1549585, clone#: C29F4, lot#: 9 -DyLight 550-conjugated mouse anti-V5 (Bio-Rad), Cat#: MCA1360D550GA, RRID: AB_2687576, clone#: SV5-Pk1 -mouse anti-Bruchpilot antibody (Developmental Studies Hybridoma Bank, nc82), Cat#: nc82, RRID: AB_2314866 -chicken anti-GFP (Abcam), Cat#: ab13970, RRID: AB_300798
Validation	Multi-colorFlip Out (MCFO) immunohistochemistry: Multi-colorFlip Out (MCFO) genetic strategy (Nern et al. 2015) uses expression of epitopes (HA, FLAG and V5) that are not endogenous to the fly genome. For MCFO immunostaining in our study we followed the exact protocol as established and validated in Drosophila by Nern et al. that uses anti-HA, anti-FLAG and anti-V5 antibodies (Nern et al. 2015). These antibodies have also each been validated prior to Nern et al: rat anti-FLAG: Manufacturer notes confirms that rat anti-FLAG (Cat#: NBP1-06712B) has also been validated as FLAG-Tag specific in Drosophila (PMID: 26573957). rabbit anti-HA: Manufacturer confirmed rabbit anti-HA antibody has Epitope tag specificity using western blot and immunohistochemical analysis comparing untransfected with HA-tag transfected COS cells (https://www.cellsignal.com/products/primary-antibodies/ha-tag-c29f4-rabbit-mab/3724#validation-data). DyLight 550-conjugated mouse anti-V5: Manufacturer notes confirm that the DyLight 550-conjugated-Mouse anti V5-Tag, clone SV5-Pk1 recognizes the sequence, IPNPLLGLD, present on the P/V proteins of the paramyxovirus, SV5 (Dunn et al.1999) and can be used to detect recombinant proteins labeled with this V5-tag (Randall et al.1993 and Zhao et al. 2005). Other immunohistochemistry looking at neuron anatomy: The anti-Bruchpilot antibody (DSHB) is the standard in the Drosophila field as a background stain that labels presynaptic active zones to provide neurophil labeling for analysis of anatomy. This antibody was originally validated for use in Drosophila to label presynaptic active zones using immunohistochemistry and to be specific to Bruchpilot protein (Wagh et al. 2006). The anti-GFP antibody (Adcam) is the standard antibody used in the field for labeling exogenous expression of Green Fluorescent Protein (GFP) in Drosophila, note that this protein is not endogenously expressed in the Drosophila genome. Manufacturer's datasheet confirm that this anti-GFP antibody has been validated using western blot and immunohistochemistry to have specificity for Green Fluorescent Protein. Manufacturer also confirms the use of this antibody for immunolabeling of GFP in Drosophila across 121 peer-reviewer manuscripts (e.g. Sykes et al. 2005 PMID: 16122730).

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	Below is a complete description from the methods sections of all transgenic Drosophila strains use in this study included how they were obtained and their citation: The following Gal4 lines were obtained from the Bloomington Drosophila Stock Center (BDSC) and are described in ref. 40: P{R60D05-Gal4}attP2, P{R60D05-lexA}attP40, P{R19C08-lexA}attP40, P{R12B01-Gal4}attP2, P{R54E12-Gal4}attP2, P{R20A02-Gal4}attP2, P{GawB}EB1 was obtained from the BDSC and is described in Wang et al. 2002 P{20XUAS-IVS-mCD8::GFP}attP40 was a gift from Barret Pfeiffer and Gerry Rubin and is described in Pfeiffer et al, 2010 P{13XLexAop2-mCD8::GFP}attP40 was obtained from the BDSC and is described in Pfeiffer et al, 2010 PBac{13xLexAop2-IVS-Syn21-Chrimson::tdT-3.1}VK00005 was a gift from Barret Pfeiffer and David Anderson and is described in Hoopfer et al. 2013 P{20X-UAS-CsChrimson-tdTomato}VK00005 was a gift from John Tuthill who obtained it from Barret Pfeiffer. P{UAS-Hsap \KCNJ2.EGFP}7 was obtained from the BDSC and is described in Hardie et al. 2001 P{UAS-GCamp6f}attP40 was obtained from the BDSC via Thomas Clandinin and is described in Chen et al. 2013 Transgenes for MultiColor FlpOut were obtained from the BDSC and are described in Nern et al. 2015 these are w[1118] P{y[+t7.7] w[+mC]=GMR57C10-FLPG5.PEST}su(Hw)attP8; PBac{y[+mDint2], and w[+mC]=10xUAS(FRT.stop)myr::smGdP-HA}VK00005 P{y[+t7.7] , and w[+mC]=10xUAS(FRT.stop)myr::smGdP-V5-THS-10xUAS(FRT.stop)myr::smGdP-FLAG}su(Hw)attP1.
Wild animals	No wild animals were used in this study.
Field-collected samples	No field samples were collected for this study.
Ethics oversight	No ethical approval was required because all experiments in this study were performed on Drosophila melanogaster.

Note that full information on the approval of the study protocol must also be provided in the manuscript.